25900528 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMOS200 - IOT FOR ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is the primary role of security frameworks in IoT? [U] / [CO1]

   a). To provide rules for device color coding

   b). To guide secure design, deployment, and operation of IoT systems

   c). To improve user interface design

   d). To reduce the number of devices in a network

2. Which of the following is a successful standard developed for IoT low-power networks? [K] / [CO2]

   a). SMTP b). POP3

   c). HTTP only d). 6LoWPAN

3. Which header must be included in Arduino IDE for ESP32 Wi-Fi? [K] / [CO3]

   a). #include <ESP32WiFi.h> b). #include <WiFi.h>

   c). #include <WiFiESP32.h> d). #include <ESP8266WiFi.h>

4. Battery life in IoT devices is especially important because: [U] / [CO1]

a). Devices are always connected to power supply

b). Devices are often deployed in remote areas

c). Devices do not consume energy

d). Devices do not require maintenance

5. Global standardization is important because IoT devices: [K] / [CO2]

a). Always use the same manufacturer

b). Need to work across borders and networks

c). Only operate locally

d). Do not require communication

6. Which device can display serial communication output from ESP32? [U] / [CO3]

a). LCD

b). Oscilloscope

c). Serial Monitor

d). Wi-Fi analyzer

7. Identify the wearable IoT application from the following. [U] / [CO1]

a). Smart glasses

b). Smart thermostat

c). Smart refrigerator

d). Smart irrigation pump

8. Which OIC design principle protects communication and data exchange? [K] / [CO2]

a). Interoperability

b). Openness

c). Security

d). Scalability

9. Which GPIO pins are input-only in ESP32? [K] / [CO3]

a). 34–39

b). 12–16

c). 21–23

d). 0–2

10. An actuator in IoT is responsible for: [K] / [CO2]

a). Storing data

b). Taking action based on received commands

c). Transmitting data to the cloud

d). Providing wireless connectivity

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11.	Discuss any three challenges related to security and cloud infrastructure in the IoT ecosystem.	[U] / [CO1]
12.	Give two examples of WoT applications in smart cities.	[U] / [CO2]
13.	Name a gas sensor used for LPG detection. Describe its features, working mechanism, and limitations.	[K] / [CO3]
14.	List and explain any three key characteristics of IoT devices.	[K] / [CO1]
15.	Define CoAP and explain why it is suitable for constrained devices.	[K] / [CO2]
16.	List three output devices commonly used with ESP32 and explain the function of each in IoT applications.	[K] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17.	Define the five key components of IoT and describe the basic function of each with relevant examples.	[K] / [CO1]
18.	Explain the objectives of OIC. How do these objectives support interoperability, openness, and security in IoT ecosystems?	[U] / [CO2]
19.	Write a program to read temperature and light values and display them on the Serial Monitor.	[U] / [CO3]
20.	Discuss the objectives and key features of IoTivity. How do they contribute to IoT standardization?	[U] / [CO2]

[10x2 = 20]